

A Full Metro-scale Time-Dependent Traffic Route Model

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ABSTRACT

We describe how to train an urban traffic model on a full metro-politan road network, up to 5-minute resolution on a week-long model, starting from GPS probes representing 5% of all traffic in that area over a 30-day period. Building such a large model (about 100,000 links, and 2016 time windows for a week), presents many challenges. First, the 5% of all traffic (about 2.5 million trips) is still incredibly sparse on this scale. This means it is essential to smooth out the data and error evaluation, and also not learn too many parameters at risk of over-fitting. Second, because the scale is so large, to be able to predict feasible routes and not just static distributions, it is important we learn a PageRank-style model. We demonstrate the model's ability for route forecasting around two changes in traffic behavior.

CCS CONCEPTS

• **Computing methodologies** → **Modeling and simulation**; Spatial and physical reasoning; • **Information systems** → *Geographic information systems*.

KEYWORDS

PageRank, road traffic, GPS trajectory, map matching, hidden Markov model, graph diffusion, urban mobility, per-link traffic estimation

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1 INTRODUCTION

A full metro-scale traffic model, we posit, should contain the following components. (1) It should consider traffic on each link of a road network. Otherwise, it is not compatible with low-level route planning, actual map-matched GPS-traces. (2) It should be temporal, with a small resolution like 5 minutes, and spanning a week. Traffic patterns are variable at this resolution level, due to commute and meal regularity. The week long models are necessary for regular changes in behavior over that period. (3) It should be able to efficiently produce actual road-network routes mirroring GPS traces, starting in each time window. These are necessary for producing realistic routes that can adapt to the environment for planning and forecasting. The efficiency is necessary so this can be deployed in other AI infrastructure for generative modeling. (4) Finally, it

should learn from and be trained on actual data so it can represent the true human behaviors and adapt if new data arises.

In this paper we demonstrate how to build such a system on the greater Salt Lake City metropolitan area. This represents a cohesive and connected ecosystem with about 100,000 links. We train this model on GPS tracing representing about 5% of the traffic for the area for a 30 day period (about 2.5 million routes). The scale of the data is large, but manageable by today's standards – the real challenges is the sparsity. After filtering and cleaning, only about 180 million GPS points remain; but at a 5-minute resolution with 2,016 time periods over a week, there are over 800 million links×time-windows to model. Even if we had a longer observation period or a higher percentage of routes, a similar challenge would remain since many routes are concentrated on busy thoroughfares and are peak times of the day. The tail of density is extremely long.

We manage to effectively build a model satisfying this desiderata through several innovations. Our baseline is a PageRank model that treats directed network links as graph vertices, and intersections model outgoing edges. We then categorize links by speed and number of lanes, and use data to learn transition probabilities between edges that favor transitions to or from higher throughput roads. However, routes typically move towards thoroughfares as they depart, and then move away from them as they reach their destination; so we design a two-phase PageRank that has separate transition probabilities as it gears up, and then speeds down. Finally, when a route ends, PageRank “teleports” to a random node to start the next walk; however, this does not reflect the pattern of route origins. So we use our observations of route starting points to also inform this teleportation probability, akin to trust-rank models []. We demonstrate that each of these steps is necessary to build an effective learned model.

Another challenge is in a trustworthy evaluation. Due to the data sparsity, even without a cross-validation split, it is not enough to directly predict the link density accuracy. Across the 4 full weeks, only 3% of links×time have at least one observation in a week, and only 1% have at least one observation in more than one week. So we disperse the sparse observations we have using graph diffusion [19]. We show how to apply this in a controlled way so that for each full week of observation, each link×time has non-zero measure, and the average relative change in value is less than the median number of difference in observations among the links×time which have observations in more than two weeks. That is, this diffusion makes the observations dense in a natural way that does not disrupt the data below the noise floor. From here, we can evaluate the model by training on one week, and measuring on the next.

Ultimately, we demonstrate an efficient, fine-grain, and full model on a metro scale data set. It produces natural temporally encoded routes, and its error is below the week-to-week variability of the diffused observations. Moreover, we demonstrate on two scenarios (a large sporting event, and an accident-induced lane closure) how we can use it to effectively forecast heavy traffic and investigate what-if scenarios.

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2 RELATED WORK

This paper designs a method to fit a calibrated model of vehicle traffic at the resolution of *every single road link* of an entire city ($\sim 100,000$ links) at 5-minute temporal resolution over a week, using probe trajectories on a fixed road network. The closest prior work falls into three groups: agent-based traffic simulators that take demand as an input rather than fit it from data [21, 22]; origin-destination estimators that operate at a coarser zone or node resolution [11, 12]; and PageRank-style random walks applied to urban networks for centrality or coarse congestion prediction [8, 23, 24].

The closest comparable effort in scope is MATSim [21, 22], a long-running agent-based simulator that, given a synthetic population, schedules, and a network, performs *forward* simulation of per-link traffic at every time of day. Our problem is the *inverse*: given observed probe trajectories, infer a per-link popularity field and validate it against held-out weeks. The two efforts complement rather than compete; we use the MATSim network model of Salt Lake City as our graph, but the calibration-from-data step is outside MATSim's scope.

A long line of transportation work estimates city-wide origin-destination (OD) matrices from cellular records combined with link counts [11] or from sparse probe trajectories combined with link counts [12]. These methods produce an OD matrix at zone or node resolution: each entry quantifies how many trips begin in zone a and end in zone b , but commits to no particular route in between. Our learned per-link traffic field is finer modeling unit than OD estimation, and also incorporates temporal modeling.

PageRank has been adapted to urban networks in a variety of ways complementary to ours. It has been used for structural centrality ranking of links in the network [8], or other ranking of urban spaces [24]. It has been used to model congestion, for instance by Wang *et.al.* [23] on a coarse 4,000 cell grid of Beijing. These methods predominantly use PageRank for analysis, and may even evaluate it against some observations, but it is not trained on full traffic observations, and not at the level of link or time granularity that we build. Our higher-resolution model, fit to real observations, will provide a better setting with which to perform these sorts of analysis.

3 DATA AND PREPROCESSING

3.1 Probe corpus

Our raw input is a corpus of sparse GPS trajectory records collected by Intrix **JEFF: get details from Cathy** in the Salt Lake City metropolitan area between August 31, 2018 and September 30, 2018, a continuous 30-day observation window. The data are distributed as a set of data rows, with ones belonging to the same device are interleaved chronologically. Each row corresponds to a single GPS fix and contains four fields:

Column	Meaning
did	anonymized device identifier (string)
ts	timestamp, formatted YYYY-MM-DD HH:MM:SS
lat	latitude of the GPS fix (decimal degrees)
lon	longitude of the GPS fix (decimal degrees)

The raw corpus contains 924,379,163 GPS locations; however roughly 80% of them fall outside the bounding box around the Salt

Lake City metro area we study (see Figure 1), and are dropped at the first preprocessing stage, leaving 179 million locations inside the metro area. After full preprocessing and the map matching of Section 3.2, the corpus is reduced to approximately 2,477,000 distinct trips, of which about 714,594 survive map matching and contribute to the population matrix. The remaining trips are discarded for one of three reasons: fewer than five GPS locations, locations that fall outside the road network, or no viable Viterbi path (from map matching). At this stage the data are purely geographic point clouds: the observations are not yet associated with any road link, and no network structure has been introduced.

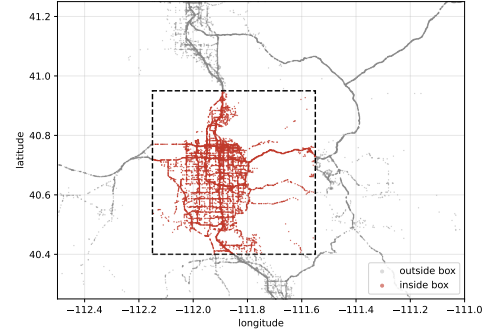


Figure 1: Raw GPS locations around the SLC bounding box (random sample of $\sim 10^5$ from 924 M). Red points fall inside the box (40.4–40.95°N, 112.15–111.55°W, dashed); gray fall outside.

3.2 Map-matching

We project the raw GPS fixes onto the road network with the Leuven implementation [1, 2] of the hidden Markov map-matcher of Newson and Krumm [3]. The matcher models the GPS-to-link perpendicular distance as a zero-mean Gaussian with standard deviation σ (in metres); σ is the only parameter and is corpus-dependent. Newson and Krumm fit $\sigma \approx 4.07$ m on their Seattle corpus; on our sparser SLC corpus we refit to be self-consistent. We re-run the matcher across a range of candidate σ (with a candidate pruning radius $\text{max_dist} = 400$ m, the matcher's default upper bound on emission distance), compute the standard deviation of the per-fix residuals over the surviving fixes, and pick the smallest σ at which the declared emission std is no longer smaller than the measured one.

Table 1: Self-consistency check for the GPS noise scale σ on the SLC corpus.

σ_{declared}	σ_{std} (filtered)	gap = $\sigma_{\text{std}} - \sigma$	verdict
1.00	2.20	+1.20	too tight
1.75	2.77	+1.02	too tight
3.00	3.15	+0.15	slightly too tight
3.50	3.22	-0.28	first $\sigma > \sigma_{\text{std}}$ (chosen)
3.75	3.25	-0.50	too loose
4.00	3.26	-0.74	too loose
5.00	3.31	-1.69	too loose

The sign of $\sigma_{\text{std}} - \sigma_{\text{declared}}$ flips between $\sigma = 3.0$ and $\sigma = 3.5$. We therefore set $\sigma = 3.5$ m throughout this paper.

Two map versions. The pipeline operates on two views of the same SLC road network. The *low-resolution* map is the MATSim simulator network ($N \approx 99,716$ directed links). The *high-resolution* map is the same OSM extract with long or curved MATSim links subdivided into shorter sub-edges so that arterial geometry is captured by collinear sub-edges. The main pipeline uses high-resolution matching aggregated to low-resolution links; the low-res baseline of Section 8.3 uses low-resolution matching instead.

3.3 Popularity matrix

The final stage of the pipeline aggregates the matched edge visits into a sparse matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ that is the input to the rest of the model. Each row corresponds to a time bin, each column to a road link, and each entry counts the number of edge visits that fell into that bin and link.

Time is partitioned into non-overlapping bins of equal width $\Delta = 300$ seconds (five minutes). For an absolute timestamp t measured in seconds since the epoch $t_0 = 2018-01-01$, the bin index is $b(t) = \left\lfloor \frac{t-t_0}{\Delta} \right\rfloor$. There are $T = 30 \times 24 \times 12 = 8,640$ five-minute bins. A five-minute resolution is chosen to balance two competing constraints: too coarse a bin (e.g., one hour) blurs the rush-hour profile of the network; too fine a bin (e.g., one minute) leaves the matrix excessively sparse, with most cells empty even on busy links.

Deduplication. The map-matched output of Section 3.2 provides, for each trip r , a list of pairs (lid_j, t_j) , one per emitting GPS location. The counting rule is:

- (1) **Bin assignment.** Each pair is assigned to the bin

$$b_j = b(t_j) = \left\lfloor \frac{t_j - t_0}{\Delta} \right\rfloor.$$

- (2) **Per-trip deduplication.** For a given trip r , all fixes that fall in the same bin and on the same link are collapsed into a single contribution. Concretely, the set of contributions from trip r is the set of distinct triples

$$\{(b_j, \text{lid}_j, r) : j = 1, \dots, M_r\}.$$

If the same trip emits five GPS points that all map to the same bin and the same link, that combination is counted exactly once.

- (3) **Accumulation.** The population-matrix entry $C_{b,i}$ is then the number of distinct trips that contributed at least one fix to bin b on link i :

$$C_{b,i} = |\{r : \exists j \text{ with } b_j = b, \text{lid}_j = i\}|. \quad (1)$$

The per-trip deduplication ensures that $C_{b,i}$ measures link occupancy by distinct vehicles in the corpus, not the GPS-emission density of those vehicles: a trip that produces several GPS points while crossing one link inside a single 5-minute window contributes one unit, not several.

Sparse matrix representation. The matrix C has dimensions $T \times N \approx 8.6 \times 10^8$ cells, where $T = 8,640$ is the number of five-minute bins (Section 3.3) and $N = 99,716$ is the number of directed road links. Only a small fraction of these cells are nonzero: at five-minute resolution most road links carry no trip in any specific window. We therefore store C in compressed sparse row format alongside the

per-row timestamps and per-column link identifiers, which on the SLC corpus reduces the artifact to roughly 7 MiB.

3.4 What C does and does not measure

The matrix C summarizes how many trips, drawn from the 714,594-trip corpus, traversed each road link in each five-minute window of the 30-day observation period. It is a direct empirical proxy for link usage and feeds the prior E_b used in the two-phase PageRank chain.

Two limitations are worth stating up front. First, the corpus is a biased sample of real traffic: it is drawn from a specific population of GPS-emitting devices and is not a uniform sample of all vehicles on the road. Links that systematically attract devices in the sample (e.g. commercial freeway corridors) are over-represented relative to links that attract a different population (e.g. residential cul-de-sacs). The matrix should therefore be read as a *prior* on link usage, not as ground-truth volume. Second, even within the corpus, sparsity is structural: at any single five-minute slot, the median number of GPS fixes per link is zero, and only a few hundred to a few thousand links carry any matched visit at all. To quantify this at the bin-of-week level, we aggregate the first four full weeks of the corpus (8,064 bins, i.e. $4 \times 7 \times 288$ five-minute slots) and ask, for each (*link, bin-of-week*) cell, in how many of the four weeks it received at least one matched fix. The result is in Table 2.

Table 2: How often each (link, bin-of-week) cell was zero across the four weeks. The cell is structurally empty if all four weeks were zero. “Active” cells are the union of $k = 0, 1, 2, 3$ rows (5.95×10^6 in total, about 3% of all cells). The total cell count 201,027,456 = $N \times 2,016$ counts (*link, bin-of-week*) pairs for a single week ($2,016 = 7 \times 288$ five-minute slots in a week); each cell carries four observations, one per calendar week.

weeks with zero count	cell count	fraction
$k = 0$ (active in every week)	304,983	0.15%
$k = 1$	499,625	0.25%
$k = 2$	1,153,583	0.57%
$k = 3$	3,994,889	1.99%
$k = 4$ (never observed in 4 weeks)	195,074,376	97.04%
total	201,027,456	100%

97% of all (link, bin-of-week) cells are zero in every one of the four weeks; only 0.15% are active in all four. Most of the matrix is therefore not just sparse but *structurally empty*, and most “active” cells are active in only one of the four weeks, i.e. represent a single noisy sample rather than a stable rate. The downstream graph-diffusion smoothing described elsewhere in the project addresses this by spreading mass to network neighbors of observed links.

4 BUILDING THE TELEPORTATION PRIOR

4.1 From counts to a prior by row-normalization

The matrix C is converted into a per-bin teleportation prior by row-normalization. For every bin b with $\sum_i C_{b,i} > 0$, define

$$E_b(i) = \frac{C_{b,i}}{\sum_j C_{b,j}}. \quad (2)$$

This makes E_b a probability distribution over the N links of the network: the share of trip-presence mass during bin b that landed on link i . Bins with no observed trips are assigned the uniform distribution $E_b(i) = \frac{1}{N}$, which acts as an uninformative fallback.

The construction of E_b is the entire purpose of the pipeline described above: every step from raw GPS in Section 3 to the sparse matrix in Section 3.3 contributes a piece of the journey from a device-level point cloud to a link-level distribution E_b that the smoothing of Section 4.2 regularizes and the model of Section 7 consumes.

4.2 Diffusion smoothing of the popularity prior

The teleportation prior E_b summarizes what was actually observed on the road network during bin b , but it has two structural shortcomings for predictive use. First, E_b is sparse: a typical 5-minute window puts non-zero mass on only a few thousand of the N links, leaving the rest of the network at exact zero. Second, the empirical signal is noisy at fine resolution: the same heavy corridor can be touched by zero observed trips in one bin and several trips in the very next, simply because no GPS device happened to emit a fix on that link inside that 5-minute slot.

We measure the scale of this noise from the four-week structure of the corpus. For each (link, bin-of-week) cell that the corpus saw at least once, we compute the standard deviation of its normalized count $E_b(i) = C_{b,i}/\sum_j C_{b,j}$ across the four weeks; this captures how much the same link in the same time-of-week jitters from week to week. We call this per-cell quantity $\tau_{b,i}$.

JEFF: State the units explicitly: $E_b(i)$ is the probability that a GPS fix falling in bin b lands on link i . The figure axis and the text should both say that.

HAMID: Concretely, $E_b(i) = C_{b,i}/\sum_j C_{b,j}$ is the fraction of all GPS fixes recorded in bin b that fell on link i , i.e. the probability that a single fix drawn uniformly from that 5-minute window lands on link i . The cross-week deviation $\tau_{b,i}$ is the spread of that probability across the four weeks at the matched bin-of-week; both quantities are dimensionless probability mass.

The median over the 5.95×10^6 active cells of Table 2 is $\tau_{\text{global}} = 2.65 \times 10^{-4}$, a single corpus-wide scalar used as a global yardstick. Differences in E_b smaller than τ are not distinguishable from week-to-week sampling noise; the smoothing step below is calibrated to this scale.

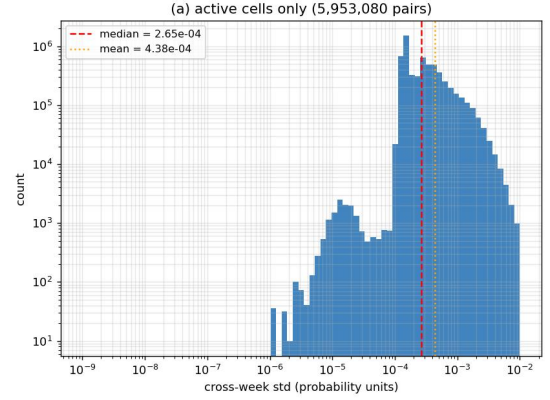


Figure 2: Distribution of the cross-week standard deviation $\tau_{b,i} = \text{std}_w E_b^{(w)}(i)$ across the four weeks, per (link, bin-of-week), restricted to the 5.95×10^6 active cells. **HAMID: Units on the horizontal axis are dimensionless probability mass: a GPS-fix-on-link probability $E_b(i) = C_{b,i}/\sum_j C_{b,j}$.**

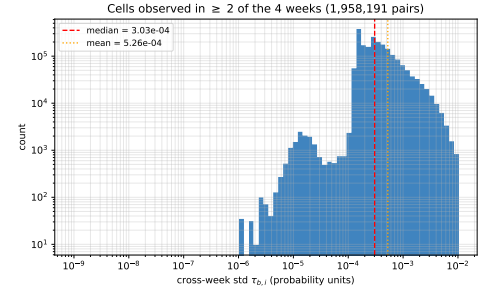


Figure 3: [temporary] Same distribution as Fig. 2 but restricted to cells observed in at least 2 of the 4 weeks (1.96×10^6 pairs). Removing the one-week-only cells (whose std collapses to a near-zero spike) exposes the actual $\tau_{b,i}$ distribution.

A graph-aware smoothing step addresses both issues at once. It lifts every link to a positive value (so the prior never has hard zeros) and borrows mass from neighboring links (so a heavy link does not disappear because of an empty bin). The smoothing is applied row by row to the count matrix $C \in \mathbb{Z}_{\geq 0}^{T \times N}$ in three steps.

Laplace pseudo-count. For every bin b and every link i , replace the raw count by

$$c_b(i) = C_{b,i} + \alpha, \quad \alpha = 10^{-2}. \quad (3)$$

After this step every entry of c_b is positive: the row is dense before any further processing.

Single-step graph diffusion. Let $P \in \mathbb{R}^{N \times N}$ be the row-stochastic adjacency matrix of the directed road graph: $P_{i,j} = \frac{1}{d_i^{\text{out}}}$ when there is a road link from i to j , and zero otherwise; here d_i^{out} is

the out-degree of link i . Apply one lazy random-walk step on the graph,

$$\tilde{c}_b = (1 - \gamma) c_b + \gamma P c_b, \quad \gamma = 0.20, \quad (4)$$

where $(P c_b)_i = \sum_j P_{i,j} c_b(j)$ is the average of $c_b(j)$ over the out-neighbors j of link i . The mass on link i is therefore averaged with the mass on the links it leads to. Finally, normalize each row to a probability distribution:

$$\tilde{E}_b(i) = \frac{\tilde{c}_b(i)}{\sum_j \tilde{c}_b(j)}. \quad (5)$$

The result $\tilde{E}_b \in \mathbb{R}^N$ is the *diffused prior*: dense, positive on every road link, and preserving the order of the heaviest links across the corpus. $\tilde{E}_b$ is used as an alternative to the raw E_b throughout the evaluation of Section 8.1.

Choosing γ from the noise floor. The diffusion strength γ is not free. We pick it so that the smoothing displaces any single cell by no more than half of that cell's own week-to-week sampling noise: any larger move and the smoother is overwriting structure with neighbor mass rather than filling in noise. Concretely, for each (link i , bin-of-week b) cell with at least one observed positive count across the four full corpus weeks $w = 1, \dots, W$, define the per-week displacement

$$D_{b,i}^{(w)}(\gamma) = \left| \tilde{E}_{b,i}^{(w)}(\gamma) - E_{b,i}^{(w)} \right|, \quad (6)$$

the cell-level displacement averaged across the four weeks

$$D_{b,i}(\gamma) = \frac{1}{W} \sum_{w=1}^W D_{b,i}^{(w)}(\gamma), \quad (7)$$

and compare it to the cross-week noise floor $\tau_{b,i}$ of the same cell. Both $D_{b,i}$ and $\tau_{b,i}$ are in probability units, so the ratio $D_{b,i}/\tau_{b,i}$ is *dimensionless*: "displacement measured in units of one noise-floor unit." The criterion is

$$D_{b,i}(\gamma) \leq \frac{1}{2} \tau_{b,i}. \quad (8)$$

We compare each cell either against its own cross-week noise scale $\tau_{b,i}$ (*per-cell criterion*, one number per active cell) or against the corpus-wide median noise scale $\tau_{\text{global}} = 2.65 \times 10^{-4}$ (*global criterion*, a single scalar shared by every cell):

$$\text{per-cell: } D_{b,i}(\gamma) \leq \frac{1}{2} \tau_{b,i}, \quad (9)$$

$$\text{global: } D_{b,i}(\gamma) \leq \frac{1}{2} \tau_{\text{global}} = 1.325 \times 10^{-4}. \quad (10)$$

On the active cells of the SLC corpus ($n_{\text{active}} = 5,954,929$, per-cell τ median $= 2.3 \times 10^{-4}$), Table 3 reports three quantiles of each ratio: the median, the 95th percentile, and the fraction of cells whose ratio exceeds the half-noise threshold.

Table 3: Quantiles of the dimensionless displacement-to-noise ratio $D_{b,i}(\gamma)/\tau$ across the 5,954,929 active SLC cells; left block uses the per-cell denominator $\tau_{b,i}$, right block uses the global denominator $\tau_{\text{global}} = 2.65 \times 10^{-4}$. Read each row as "at this γ , half / 95% / fraction-above-0.5 of active cells are displaced by at most so-many noise units." The criterion is the largest γ whose 95th-percentile ratio stays ≤ 0.5 . **JEFF: remove some of the columns**

γ	per-cell criterion			global criterion		
	med.	95th pct.	> 0.5	med.	95th pct.	> 0.5
0.00	0.000	0.000	0.00%	0.000	0.000	0.00%
0.05	0.038	0.100	0.42%	0.034	0.215	0.90%
0.10	0.069	0.186	1.00%	0.062	0.393	3.31%
0.12	0.081	0.219	1.27%	0.073	0.463	4.40%
0.13	0.087	0.236	1.41%	0.079	0.498	4.96%
0.14	0.093	0.253	1.56%	0.084	0.533	5.53%
0.15	0.099	0.270	1.73%	0.090	0.568	6.10%
0.20	0.130	0.354	2.62%	0.117	0.741	8.95%
0.25	0.160	0.436	3.63%	0.144	0.912	11.87%
0.30	0.190	0.518	5.39%	0.171	1.081	14.81%
0.40	0.248	0.680	9.42%	0.224	1.413	20.64%
0.50	0.306	0.838	17.29%	0.276	1.737	26.59%
0.70	0.418	1.145	32.10%	0.376	2.364	37.83%

Table 4: [temporary] Displacement of zero cells ($n_{\text{zero}} = 1.95 \times 10^8$) in units of $\tau_{\text{global}} = 2.65 \times 10^{-4}$.

γ	median	95th pct.	> 0.5
0.00	0.0004	0.0004	0.00%
0.05	0.0004	0.0004	0.01%
0.10	0.0004	0.0004	0.03%
0.12	0.0004	0.0004	0.04%
0.13	0.0004	0.0004	0.04%
0.14	0.0004	0.0004	0.05%
0.15	0.0004	0.0004	0.05%
0.20	0.0004	0.0004	0.08%
0.25	0.0004	0.0004	0.10%
0.30	0.0004	0.0004	0.13%
0.40	0.0004	0.0004	0.20%
0.50	0.0004	0.0004	0.28%
0.70	0.0003	0.0004	0.46%

Reading the zero-cell table. The median and 95th-percentile ratios are essentially constant at $\approx 4 \times 10^{-4}$ across all γ : at least 95% of zero cells never see diffusion mass, they only carry the Laplace floor α/N . The fraction of zero cells exceeding the half-noise tolerance is 0.08% at $\gamma = 0.20$ ($\sim 1.5 \times 10^5$ cells out of 1.95×10^8); these are the cells immediately adjacent to active links and they are the ones that get visibly filled in. The takeaway is that the diffusion is bounded even on the cells where the per-cell criterion of Table 3 is undefined: it lifts a small adjacency-defined tail of zero cells into the noise band and leaves the rest at the Laplace floor.

Choice of γ . Under the per-cell criterion the 95th-percentile ratio first crosses 0.5 between $\gamma = 0.25$ (0.436) and $\gamma = 0.30$ (0.518), so

$\gamma_{\max}^{\text{pc}} \approx 0.25$. The stricter global criterion crosses between $\gamma = 0.13$ (0.498) and $\gamma = 0.14$ (0.533), giving $\gamma_{\max}^{\text{global}} \approx 0.13$. We use $\gamma = 0.20$ throughout the paper: it sits comfortably inside the per-cell ceiling, the median active cell is displaced by only $0.117 \tau_{\text{global}}$ even under the global criterion, and the 95th-percentile ratio crosses the half-noise bound only mildly (0.741).

Violator tail at $\gamma = 0.20$. To see how far past the tolerance the violators sit, we sweep the threshold η in $D_{b,i}(0.20) > \eta \tau_{\text{global}}$ across the active cells:

η	0.2	0.5	0.8	1.0	1.5	2.0
fraction $> \eta \tau_{\text{global}}$	28.5%	8.9%	4.4%	3.0%	1.3%	0.6%

The violator tail thins quickly: of the 8.9% that exceed half the noise floor, only 3% exceed τ_{global} and fewer than 0.7% exceed $2 \tau_{\text{global}}$, so the smoothing rarely changes a cell by more than its own week-to-week jitter. The per-cell and global criteria agree on the median and differ only on the upper tail (the per-cell version self-normalizes against busy cells, the global version measures every cell against the corpus median); we report both and use $\gamma = 0.20$ as the working compromise.

5 EVALUATION FRAMEWORK

We now fix the metrics and protocol used by every subsequent evaluation, including the single-phase baseline of Section 6 and the two-phase chain of Section 7.

5.1 Error measures

To evaluate a chain we compare two probability vectors over the N road links: a *truth* vector t and a *prediction* vector p , both summing to one (each entry is the share of trip-presence mass on one link in a given bin). Two scalar summaries are used.

Mean Relative Error (MRE). For a numerical regularizer $\varepsilon \geq 0$, define the per-link relative error

$$r_i = \frac{|t_i - p_i|}{t_i + \varepsilon}, \quad (11)$$

which is dimensionless: numerator and denominator are both in probability units, so r_i measures error as a fraction of the truth itself. We aggregate r_i across every link in the network:

$$\text{MRE}_{\text{overall}} = \frac{1}{N} \sum_{i=1}^N r_i. \quad (12)$$

The regularizer ε is needed only when t has links with $t_i = 0$: it prevents division by zero and acts as a numerical floor that controls how much empty-truth links inflate the average. We report $\varepsilon = 10^{-6}$ throughout, and additionally $\varepsilon = 0$ in configurations where the diffused prior $\tilde{E}_b$ guarantees $t_i > 0$ on every link.

HAMID: To anchor the scale, Table 5 summarizes the number of matched GPS fixes in each 5-minute bin across the 8,640 bins of the corpus. The busiest bin contains $M_{\max} = 20,881$ fixes, so the smallest possible nonzero per-link probability anywhere in the corpus is $1/M_{\max} \approx 4.8 \times 10^{-5}$. With the uniform per-link probability $1/N \approx 1.00 \times 10^{-5}$ and the empirical noise floor $\tau_{\text{global}} = 2.65 \times 10^{-4}$, our $\varepsilon = 10^{-6}$ sits below all three: it stabilizes the denominator for links

with $t_i = 0$ without otherwise perturbing the links we care about. **JEFF:** why $\varepsilon = 10^{-6}$

Table 5: Matched GPS fixes per 5-minute bin across the 8,640 bins of the corpus.

statistic	value
total matched fixes	58,410,362
min	295
10th percentile	1,405
median	4,823
mean	6,760
90th percentile	14,746
max	20,881

Three comparisons. For every target bin t in the evaluation sample we compute the chain’s prediction $\hat{v}(t)$ by running the iteration (19) until convergence with $E_b(t)$ (or $\tilde{E}_b(t)$) as the teleportation prior. We then form three (truth, prediction) pairs:

- (1) **dataset** $\rightarrow$ **dataset** (week to next week): truth = $E_b(t + 7 \text{ days})$, prediction = $E_b(t)$. Reports the natural week-to-week variability of the empirical signal at the same weekday and hour-of-day; serves as the *noise floor* the model needs to beat.
- (2) **dataset** $\rightarrow$ **model** (forecast next week): truth = $E_b(t + 7 \text{ days})$, prediction = $\hat{v}(t)$. Reports how well the chain, fed this week’s prior, predicts next week’s empirical signal.

With the diffused prior, the same two comparisons replace E_b by $\tilde{E}_b$ on *both sides*: the truth vector, the prediction vector when the prediction is itself a prior (comparison 1), and the chain input that produces $\hat{v}(t)$ (comparison 2) are all diffused before the MRE is taken. This keeps every comparison apples-to-apples within a single choice of prior.

Sampling protocol. For each of the seven days of the week (Mon, ..., Sun) we draw 120 source bins uniformly at random, for a total of $7 \times 120 = 840$ target bins. A bin is eligible only if both the bin itself and its +7-day sibling exist in the matrix. In our 30-day corpus this restricts source bins to days 1–23 (Sept 1 to Sept 23) paired with siblings in days 8–30 (Sept 8 to Sept 30); the source bins therefore span weeks 1–3 and the first days of week 4, not just the first week. The seed is fixed (7), so the sample is identical across configurations.

6 BASELINE: SINGLE-PHASE PAGERANK

The most natural baseline on the popularity prior E_b is the single-phase PageRank chain with E_b as the teleportation distribution. Given the row-stochastic adjacency P of the directed road graph (each row of P being uniform over the outgoing road neighbors of one link) and a teleport probability $p \in (0, 1)$, the iteration is

$$v_{n+1} = p E_b + (1 - p) P^\top v_n, \quad (13)$$

applied until $\|v_{n+1} - v_n\|_1 < 10^{-7}$, with a safety cap of 300 iterations. The cap is conservative: the power iteration converges geometrically at rate $1 - p$, so the theoretical worst-case number of iterations

to reach L1 error 10^{-7} is

$$n^* = \frac{\ln 10^{-7}}{\ln(1-p)} = \frac{\ln 10^{-7}}{\ln 0.9492} \approx 310, \quad (14)$$

which sits just above 300. We checked empirically on 1,500+ random target bins and the chain converged before 300 iterations in every single case; the observed maximum was 254 iterations and the median was 209. The cap is therefore never hit in practice and the recorded results are at full 10^{-7} tolerance throughout. At every step the walker teleports back to the prior with probability p and otherwise takes a uniform random step along an outgoing road edge. The stationary distribution v^* of (13) is the single-phase prediction for bin b .

Calibrating p from the trip corpus. The chain in (13) is a geometric absorption process: the number of graph steps before a teleport is $\text{Geom}(p)$ on $\{0, 1, 2, \dots\}$ with mean $\frac{1-p}{p}$. We match p to the empirical mean trip length $\mathbb{E}[K_{\text{total}}]$ by method-of-moments, exactly as the two-phase chain matches β and ρ to the per-phase lengths:

$$p = \frac{1}{1 + \mathbb{E}[K_{\text{total}}]}. \quad (15)$$

On the noise=3.5 matched corpus (714,594 trips, all route lengths), the empirical mean is $\mathbb{E}[K_{\text{total}}] = 18.68$, which gives $p = 0.0508$. The chain has only this one free parameter; the “damping factor” $1-p = 0.9492$ that some authors report is just a relabeling. Figure 4 shows the empirical K_{total} histogram with the $\text{Geom}(p = 0.0508)$ reference overlaid.

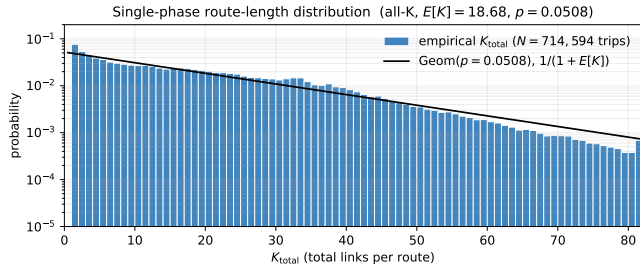


Figure 4: Empirical histogram of trip length K_{total} on the 714,594-trip corpus, with the matched $\text{Geom}(p = 0.0508)$ overlay.

Why this is a fair baseline. Three reasons. First, single-phase PageRank is the standard graph-walk model on top of which the two-phase chain is built, so the two share a graph, a prior E_b , and a power-iteration solver; the only structural difference is the absence (single-phase) or presence (two-phase) of an up→down split. Second, the teleport parameter p is calibrated from the same trip corpus by the same method-of-moments estimator that yields β and ρ , so the comparison is not penalized by an artificially detuned baseline. Third, the baseline runs on the same 840 target bins, the same metric, and the same three comparisons as the two-phase chain, per the evaluation contract.

6.1 Why single-phase fails on our data

Evaluating (13) on the same 840 target bins as Table 8 (120 per weekday, seed 7, all three eps configurations) gives the numbers in Table 6.

Table 6: Single-phase PageRank baseline (chain (13) with $p = 0.0508$) on 840 target bins (120 per weekday, seed 7). Conventions match Table 8.

Comparison	No diffusion $\varepsilon = 10^{-6}$	Diffused $\varepsilon = 10^{-6}$	Diffused $\varepsilon = 0$
1. dataset→dataset (next week)	5.82	0.68	0.90
2. dataset→model (forecast)	8.70	1.07	1.46

Two readings of Table 6, paired against the two-phase numbers in Table 8.

The forecast loses to the noise floor. Forecast Overall MRE is 1.07 against a noise floor of 0.68 (diffused, $\varepsilon = 10^{-6}$). The single-phase chain therefore fails the contract’s pass criterion that forecast must not be worse than last-week-at-the-same-slot. The two-phase chain in Table 8 reaches forecast Overall MRE of 1.04 on the same comparison, only marginally better than the single-phase baseline.

Why the single-phase mixes too much. At $p = 0.0508$ the chain takes on average $\frac{1}{p} = 19.7$ graph steps before teleporting back to E_b , which is the empirical mean trip length, so the chain is calibrated to the right time scale. But 19.7 uniform random steps on the SLC road graph are enough to delocalize the walker’s distribution across the network: the steady state of a random walk on a connected graph is the degree-proportional distribution, and even at finite damping the chain has visibly relaxed away from E_b on the heavy links. The two-phase split absorbs that mixing into the down phase ($\rho = 0.101$ leak) while keeping the up phase close to E_b ($\beta = 0.102$ leak, applied on $P^\uparrow$ rather than the uniform P), which is why the two-phase chain in Table 8 preserves the heavy-traffic head where the single-phase baseline does not. The next section formalizes this two-phase chain.

7 TWO-PHASE PAGERANK CHAIN

The two-phase chain of this section turns a popularity prior (the raw E_b from Section 3.3 or the diffused $\tilde{E}_b$ from Section 4.2) into a per-link prediction by running a discrete-time Markov chain on a road-network state space. Its design departs from a single-phase chain (one state per link) in one important respect: each road link $i \in \{1, \dots, N\}$ is represented by *two* states, an *up-phase* state $i^\uparrow$ and a *down-phase* state $i^\downarrow$, so the chain has $2N$ states in total. The doubled state space lets the chain track two distinct regimes that a vehicle on the network can occupy, and lets each regime contribute its own characteristic mass to the link-level prediction.

7.1 What the two phases represent

The two phases of the chain correspond to two qualitatively different regimes of a vehicle’s behavior on the road network:

- The *up phase* represents a vehicle that has just appeared in the network and is choosing where to go: it has not yet committed to traveling along a specific corridor. Steps within the up phase let mass redistribute along the graph before any actual travel begins.
- The *down phase* represents a vehicle that is *traveling* along the network, advancing from one link to the next according to road geometry. Steps within the down phase trace out the kind of multi-link path that a real trip would follow.

A single-phase Markov chain on the N physical links cannot distinguish these two regimes: it would force a single transition rule to govern both “where does mass come from” and “where does mass travel to”, and any stationary distribution it produced would fold the two contributions into one number per link. The two-phase chain keeps them separate, so that the steady-state mass on each physical link decomposes into a contribution from up-phase deliberation and a contribution from down-phase travel, both controlled by their own transition rule. The next subsections specify those transition rules and the parameters that govern them.

7.2 Transition rule

The state space is $\{1, \dots, N\} \times \{\uparrow, \downarrow\}$: every road link i is duplicated into an up-state $i^\uparrow$ and a down-state $i^\downarrow$. A mass vector on this space is a probability over $2N$ entries (it sums to one), and the chain’s stationary distribution decomposes naturally into an up-phase share $v^\uparrow(i)$ and a down-phase share $v^\downarrow(i)$ per link.

The chain is fully specified by the rule that takes a current state to its next state. From an up-phase state $i^\uparrow$ the rule is

- with probability $1 - \beta$, the chain stays in the up phase and moves to one of i ’s graph neighbors $j^\uparrow$, with the choice governed by an up-phase transition matrix $P^\uparrow \in \mathbb{R}^{N \times N}$;
- with probability $\beta \in (0, 1)$, the chain commits to traveling along link i and moves to the down-phase state $i^\downarrow$ on the same link.

From a down-phase state $i^\downarrow$ the rule is

- with probability $1 - \rho$, the chain stays in the down phase and moves to a graph neighbor $j^\downarrow$, governed by a down-phase transition matrix $P^\downarrow \in \mathbb{R}^{N \times N}$;
- with probability $\rho \in (0, 1)$, the trip ends and the chain teleports back to a fresh up-phase state drawn from the prior E_b : it jumps to $j^\uparrow$ with probability $E_b(j)$.

We adopt the *column-stochastic* convention throughout: column i of any of these matrices stores the probability distribution over next states, given that the chain is currently in state i . In particular, $P^\uparrow$ and $P^\downarrow$ are column-stochastic matrices supported on the directed road graph: the entry $P_{j,i}^\uparrow$ is the probability of moving to link j from link i in the up phase, and is zero whenever there is no road link from i to j (and similarly for $P^\downarrow$). Below, v denotes the chain’s mass vector at iteration step n ; like E_b , it is a probability over the $2N$ states (so $\sum_i v^\uparrow(i) + \sum_i v^\downarrow(i) = 1$). The chain’s prediction for link i is the sum $v^\uparrow(i) + v^\downarrow(i)$, itself a probability over the N links.

The explicit construction of $P^\uparrow, P^\downarrow$ is as follows. Each link ℓ in the network carries two physical attributes from the graph: a free flow speed $s(\ell)$ and a lane count $\lambda(\ell)$. Let $\bar{s}(\ell) = s(\ell)/\bar{s}$ and

$\bar{\lambda}(\ell) = \lambda(\ell)/\bar{\lambda}$ denote these attributes normalized by their per-link mean over the network, and define the per-link physics weight

$$w^\uparrow(\ell) = \bar{s}(\ell)^{\alpha_s} \bar{\lambda}(\ell)^{\alpha_l}, \quad w^\downarrow(\ell) = \bar{s}(\ell)^{-\alpha_s} \bar{\lambda}(\ell)^{-\alpha_l}, \quad (16)$$

with two real exponents $\alpha_s, \alpha_l \in \mathbb{R}$ that parametrize how strongly the chain favors fast or wide links in each phase. Each kernel is then a w -weighted row-stochastic walk on the directed road graph, made column-stochastic by transpose:

$$P_{j,i}^\uparrow = \frac{w^\uparrow(j) \mathbf{1}[i \rightarrow j]}{\sum_{j': i \rightarrow j'} w^\uparrow(j')}, \quad (17)$$

where $i \rightarrow j$ denotes “there is a directed road link from i to j ”; $P_{j,i}^\downarrow$ is defined identically with $w^\downarrow$ in place of $w^\uparrow$. The down-phase weight is the up-phase weight with (α_s, α_l) sign-flipped, so the two phases are mirrored: when the up phase prefers slow or narrow links the down phase prefers fast or wide links, and vice versa. The default chain used throughout the evaluation has $\alpha_s = \alpha_l = 0$, which makes $w^\uparrow, w^\downarrow \equiv 1$ and both kernels reduce to the uniform-over-out-neighbors walk on the directed road graph; the speed/lane exponents come back as tuning knobs in the intervention experiment of Section 8.4.

Equivalently, ordering the $2N$ states as $(1^\uparrow, \dots, N^\uparrow, 1^\downarrow, \dots, N^\downarrow)$, the transition matrix of the chain is the column-stochastic $2N \times 2N$ block matrix

$$M_b = \begin{pmatrix} (1 - \beta) P^\uparrow & \rho E_b \mathbf{1}^\top \\ \beta I & (1 - \rho) P^\downarrow \end{pmatrix}, \quad (18)$$

where I is the $N \times N$ identity, $\mathbf{1} \in \mathbb{R}^N$ is the all-ones column vector, and $E_b \in \mathbb{R}^N$ is the teleportation prior treated as a column vector. The rank-1 block $\rho E_b \mathbf{1}^\top$ encodes the “every down-state gives mass back to the up phase via the same prior E_b ” rule: each of its columns is the column vector ρE_b . With this convention the chain’s mass distribution $v \in \mathbb{R}^{2N}$ evolves as the simple matrix-vector product

$$v_{n+1} = M_b v_n, \quad (19)$$

with no transpose required at any step of the iteration.

The two scalars β and ρ control how long the chain stays in each phase before switching. A small β lets the up-phase walk redistribute mass over many graph neighbors before committing to travel, while a small ρ lets a single down-phase trip extend over many links before restarting. The values used in practice are not free parameters: they are method-of-moments estimates from the trip corpus, $\beta = \frac{1}{1 + \mathbb{E}[L_\uparrow]}$ and $\rho = \frac{1}{1 + \mathbb{E}[L_\downarrow]}$, where $L_\uparrow$ and $L_\downarrow$ are the empirical up- and down-phase clock lengths of the matched corpus. Their derivation from the raw matched routes is the subject of Section 7.3.

7.3 Calibrating β and ρ from the trip corpus

The two scalars are not tuned against the evaluation metric; they are read off the empirical distribution of trip lengths in the matched-route corpus, in three steps.

Per-trip peak split. Every matched route is a sequence of road links $(\ell_1, \ell_2, \dots, \ell_K)$ produced by the Viterbi pass of Section 3.2. For each trip we identify a single *peak link* ℓ_p as the link with the largest score $s(\ell) = \text{freespeed}(\ell) \times \text{numLanes}(\ell)$, with ties broken by the link closest to the trip’s middle. The peak link plays the role of

the trip’s fastest, highest-capacity segment, the freeway core of a typical commute. The two phases of the chain are then defined as the prefix and the suffix of the route relative to this peak,

$$L_{\uparrow} = |\{j : j < p\}|, \quad L_{\downarrow} = |\{j : j > p\}|, \quad (20)$$

so that $L_{\uparrow}$ counts the links traversed before the peak (exclusive), $L_{\downarrow}$ counts the links traversed after the peak (exclusive), and the peak link itself contributes the single intermediate transition $i^{\uparrow} \rightarrow i^{\downarrow}$ of Section 7.2; in particular, $L_{\uparrow} + 1 + L_{\downarrow} = K$. By construction $L_{\uparrow}, L_{\downarrow} \in \{0, 1, 2, \dots\}$: a route can begin or end on its own peak link, in which case the corresponding phase has length zero. Short trips ($K = 1, 2$) are handled by the convention $p = 0$ (the trip’s first link is its own peak), which yields $L_{\uparrow} = 0$ and $L_{\downarrow} = K - 1$ and contributes naturally to the empirical distributions. All 714,594 matched routes are therefore included in the calibration sample.

Empirical distribution and method-of-moments. Figure 5 shows the histograms of $L_{\uparrow}$ and $L_{\downarrow}$ on this sample, with a geometric reference curve overlaid in black on each panel. Both empirical distributions are heavy at zero and decay roughly exponentially, which is the signature of a geometric distribution on $\{0, 1, 2, \dots\}$ with success probability p and mean $\frac{1-p}{p}$. The black curves overlay $\text{Geom}(p)$ with p matched to the empirical mean through

$$\mathbb{E}[L] = \frac{1-p}{p} \iff p = \frac{1}{1 + \mathbb{E}[L]}, \quad (21)$$

which is the method-of-moments estimator for a geometric on $\{0, 1, \dots\}$. Read each histogram bin as “in this many trips the prefix (resp. suffix) was exactly L links long.” Setting $\beta = 1/(1 + \mathbb{E}[L_{\uparrow}])$ means the chain’s average up-phase clock length matches the empirical mean, and likewise for ρ ; the values $\beta, \rho \approx 0.10$ correspond to chains that stay in each phase about ten steps on average. Reading the empirical means off the histograms,

$$\mathbb{E}[L_{\uparrow}] = 8.81, \quad \mathbb{E}[L_{\downarrow}] = 8.87, \quad (22)$$

and applying (21) to each phase yields

$$\beta = \frac{1}{1 + \mathbb{E}[L_{\uparrow}]} = 0.102, \quad \rho = \frac{1}{1 + \mathbb{E}[L_{\downarrow}]} = 0.101. \quad (23)$$

These are the values fed into the chain in Sections 7.2 and 8.1; they are not chosen to fit the evaluation, only to match the average up- and down-phase clock lengths of the matched corpus. Both panels of Fig. 5 are well described by their geometric reference in the body across the full range, with the empirical bars sitting close to the geometric line throughout (a small under-dispersion at large L shows up as the bars being slightly below the reference for $L_{\uparrow}, L_{\downarrow} \gtrsim 30$).

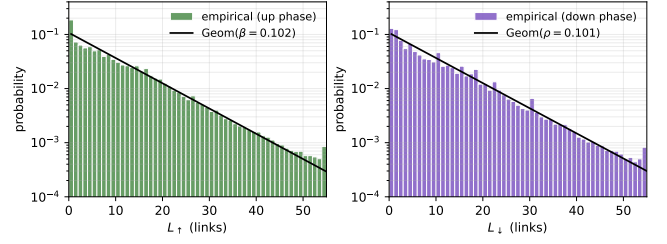


Figure 5: Empirical histograms of $L_{\uparrow}$ (left) and $L_{\downarrow}$ (right) on the full 714,594-trip matched corpus, with the matched geometric overlay in black.

Why the geometric is the right reference. In the chain of Section 7.2, the number of up-phase steps before the chain commits to travel is, by construction, a geometric random variable with success probability β on the support $\{0, 1, 2, \dots\}$, and the number of down-phase steps before the chain restarts is geometric with probability ρ on the same support. Calibrating β and ρ by the method-of-moments (21) therefore guarantees that the chain matches the empirical mean phase length exactly. The geometric assumption is approximate (the empirical tails are slightly heavier than geometric, as visible in Fig. 5), but the two means are by construction correct, which is the moment that the chain consumes.

Quantifying the fit. We pair the visual fit of Fig. 5 with the Kolmogorov-Smirnov statistic $D = \sup_k |F_{\text{emp}}(k) - F_{\text{geom}}(k; p)|$, where

$$F_{\text{emp}}(k) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[X_i \leq k]$$

is the empirical CDF over the n observed phase lengths $\{X_i\}_{i=1}^n$,

$$F_{\text{geom}}(k; p) = 1 - (1 - p)^{k+1}, \quad k \in \{0, 1, 2, \dots\},$$

is the CDF of the fitted geometric, and

$$p = \frac{1}{1 + \mathbb{E}[L]}$$

is its maximum-likelihood estimator. Smaller D means a tighter empirical match. Table 7 reports D in three configurations. Column D_{single} fits one geometric to the full route length $K = L_{\uparrow} + 1 + L_{\downarrow}$; columns $D_{\uparrow}$ and $D_{\downarrow}$ fit geometrics to $L_{\uparrow}$ and $L_{\downarrow}$ separately. The first row matches on the low-resolution map and aggregates to low-resolution links; the second matches on the high-resolution map and aggregates to the same links; the third both matches and aggregates at high-resolution.

Table 7: Kolmogorov-Smirnov distance D between empirical phase lengths and the fitted geometric. Lower is better; read each row as “fitting a $\text{Geom}(p)$ to the $L_{\uparrow}$ (resp. $L_{\downarrow}$) histogram leaves a D of this size under K-S.”

Matcher	Granularity	n	D_{single}	$D_{\uparrow}$	$D_{\downarrow}$
low-res	low-res links	293,066	0.188	0.129	0.103
high-res	low-res links	621,986	0.129	0.030	0.032
high-res	high-res sub-edges	306,659	0.102	0.138	0.148

Two findings follow. On the low-resolution aggregation that the chain runs on, splitting the route at the freespeed $\times$ permlanes peak reduces D by 76% relative to the single-geometric fit, and the matching change from the low-resolution to the high-resolution map tightens $D_{\uparrow}$ and $D_{\downarrow}$ by a further 76% and 69% on the same granularity. The empirical variance moves the same way: the overdispersion ratio $\frac{\text{Var}(L_{\uparrow})}{\mathbb{E}[L_{\uparrow}](1+\mathbb{E}[L_{\uparrow}])}$ moves from 2.07 on the low-resolution map to 0.93 on the high-resolution map, sitting just below unity (slight under-dispersion) and well inside the geometric model's natural variance range.

At high-resolution sub-edge granularity, neither single nor split geometric is a good fit; the empirical variance is overdispersed by a factor of 1.8 to 2.0 and the up versus down split, computed at low-resolution before being projected to sub-edges, does not separate the empirical distribution cleanly. We therefore keep the geometric calibration at low-resolution ($\beta = 0.102$, $\rho = 0.101$) and treat the high-resolution view of the same data as a descriptive aggregation only.

8 EXPERIMENTS

The evaluation framework (Section 5) defines the metric (Overall MRE), the two comparisons (dataset-to-dataset, forecast), and the 840-bin sampling protocol used by every table below.

8.1 Evaluation results

Table 8 reports the three comparisons in three configurations: raw prior with $\varepsilon = 10^{-6}$, diffused prior with $\varepsilon = 10^{-6}$, and diffused prior with $\varepsilon = 0$. The chain uses the calibrated parameters $\beta = 0.102$, $\rho = 0.101$ and $P^{\uparrow}, P^{\downarrow}$ uniform over outgoing road-graph neighbors. Iteration tolerance is 10^{-7} , maximum 300 iterations.

Table 8: Overall MRE of the two-phase chain across 840 random target bins (120 per day of the week) on the SLC corpus. $\beta = 0.102$, $\rho = 0.101$. Lower is better. “No diffusion” uses the raw prior E_b of Section 3.3; “Diffused” uses the diffused prior $\tilde{E}_b$ of Section 4.2.

Comparison	No diffusion $\varepsilon = 10^{-6}$	Diffused $\varepsilon = 10^{-6}$	Diffused $\varepsilon = 0$
1. dataset $\rightarrow$ dataset (next week)	5.82	0.68	0.90
2. dataset $\rightarrow$ model (forecast)	8.53	1.04	1.42

Two readings of the table.

Forecast sits above the noise floor. Row 2 forecast Overall MRE is 1.04, above the noise floor of row 1 at 0.68. The chain's forecast of next week's signal is therefore worse than last-week-at-the-same-slot. Since the chain has no temporal dynamics (no time index in M_b except through the prior of the target bin), improving forecast performance would require additional signal: weekly periodicity, multi-bin context, or external covariates, none of which are part of the current model.

Empty-link contribution to the overall MRE. The Overall MRE without the regularizer inflates to ~ 8 in the leftmost column because the many empty links contribute $\frac{|p_i|}{\varepsilon}$ each; $\varepsilon = 10^{-6}$ caps this contribution but does not eliminate it. With every truth value positive, the Overall MRE drops to 0.67–1.37, a range in which the metric is interpretable as a relative error average.

8.2 Additive ladder: damping and road-type tuning on the whole network

This subsection isolates the contributions of the damping rate and the road-type weighting in two stages on the same evaluation sample. The sample is a set of 840 calendar pairs (t , $t+7$ d), drawn so that exactly 120 pairs sit on each day of the week (seed 7), with t chosen from the first 23 days of the corpus so that the next-week sibling is in range. The model receives $\tilde{E}_b(t)$ and predicts $\tilde{E}_b(t+7$ d). The per-pair error is the Overall MRE (over all N links), and we report its mean across the 840 pairs. Parameter bounds follow the geometric-fit discussion above: α , β , $\rho \in [0.01, 0.15]$, while α_s and α_l are unconstrained.

Damping sweep (no road-type tuning). We first hold $\alpha_s = \alpha_l = 0$ and sweep the damping rate. For the single-phase chain the rate is α . For the two-phase chain we set $\beta = \rho$ and sweep the common value, so that the two columns differ only by the chain structure.

Table 9: Damping sweep on the whole network, 840 pairs (t , $t+7$ d), no road-type weighting. Overall MRE (lower is better); bold marks the best in each column.

rate	Single-phase (α)	Two-phase ($\beta = \rho$)
0.03	1.130	1.181
0.04	1.098	1.154
0.05	1.072	1.131
0.06	1.049	1.110
0.07	1.029	1.091
0.15	0.923	0.984

Both columns improve monotonically as the damping rate increases toward the upper bound of the search box. Single-phase outperforms two-phase at every common rate by approximately 0.05–0.06 on Overall MRE.

Road-type tuning on top. Table 10 fixes the damping rate at the upper bound and adds road-type weighting, either lanes only, speed only, or both jointly. Each tuned operating point is found by L-BFGS-B with finite-difference gradients and five random starts; the optimization minimizes Overall MRE on the same 840 pairs. The vanilla rows from Table 9 are reproduced for reference at the same damping rate.

Table 10: Road-type tuning on the 840 pairs, Overall MRE objective. Single-phase block: $\alpha \in [0.01, 0.15]$, α_s, α_l **unbounded unless noted. Two-phase block:** $\beta, \rho \in [0.01, 0.15]$, α_s, α_l **unbounded unless noted. Vanilla rows are objective-independent; tuned rows report the operating point at the optimizer’s solution.**

variant	parameters	Ov. MRE
<i>Single-phase</i>		
uniform prior	$\alpha = 0.15, E_b = 1/N$	1.309
vanilla	$\alpha = 0.05$	1.072
vanilla	$\alpha = 0.15$	0.923
+ lanes only	$\alpha = 0.15, \alpha_l = +0.337$	0.916
+ speed only	$\alpha = 0.15, \alpha_s = +2.859$	0.870
+ length only	$\alpha = 0.15, \alpha_{len} = +0.001$	0.923
+ both	$\alpha = 0.15, \alpha_s = +2.674, \alpha_l = +0.169$	0.869
+ both, with len	$\alpha = 0.15, \alpha_s = +2.677, \alpha_l = +0.167, \alpha_{len} = +0.009$	0.869
+ both, α fixed	$\alpha = 0.05; \alpha_s = +2.774, \alpha_l = +0.243$	1.002
<i>Two-phase</i>		
uniform prior	$\beta = \rho = 0.15, E_b = 1/N$	1.311
vanilla	$\beta = 0.102, \rho = 0.101$	1.041
vanilla	$\beta = \rho = 0.15$	0.984
+ lanes only	$\beta = 0.15, \rho = 0.15, \alpha_l = +1.875$	0.962
+ speed only	$\beta = 0.15, \rho = 0.15, \alpha_s = +5.136$	0.930
+ both	$\beta = 0.15, \rho = 0.15, \alpha_s = +4.621, \alpha_l = +0.458$	0.929
+ both, (β, ρ) fixed	$(\beta, \rho) = (0.102, 0.101); \alpha_s = +2.816, \alpha_l = +0.209$	0.970
<i>Two-phase, classical form (appendix)</i>		
uniform prior	$\alpha = 0.15, \beta = 0.15, E_b = 1/N$	1.309
vanilla	$\alpha = 0.05, \beta = 0.15$	1.058
vanilla	$\alpha = 0.15, \beta = 0.15$	0.908
+ lanes only	$\alpha = 0.15, \beta = 0.15, \alpha_l = +0.340$	0.901
+ speed only	$\alpha = 0.15, \beta = 0.15, \alpha_s = +2.839$	0.857
+ length only	$\alpha = 0.15, \beta = 0.15, \alpha_{len} = +0.001$	0.908
+ both	$\alpha = 0.15, \beta = 0.15, \alpha_s = +2.655, \alpha_l = +0.167$	0.856
+ both, with len	$\alpha = 0.15, \beta = 0.15, \alpha_s = ?, \alpha_l = ?, \alpha_{len} = ?$	TBD
+ both, (α, β) fixed	$(\alpha, \beta) = (0.0508, 0.102); \alpha_s = +2.830, \alpha_l = +0.237$	0.984

Discussion. HAMID: Collapsing the chain into a single block matrix (Eq. (18)) loses to single-phase PageRank on the full-network week-to-week prediction in Table 10. The two models are not nested: single-phase is not recoverable as a special case of this two-phase

form, so the loss is not an inference artefact but a structural one. The two-phase form only outperforms single-phase on the Sept-15 stadium cluster (Table 11), not on the whole network. The classical-form two-phase chain in Appendix A, where the transition matrix and the teleportation vector are kept separate, leads both single-phase and the block-matrix two-phase on every ladder rung in our early runs. A second observation: every chain underperforms when α, β, ρ are pinned to the geometric-fit values of Section 7.3; raising the teleportation coefficient toward the upper bound 0.15 improves the prediction because more mass is placed on the prior E_b , and the prior, which is the same bin one week earlier after γ -diffusion, is

itself a strong predictor of the target bin.

8.3 Baselines and alternative pipelines

To show that our model is doing real work, we compare it against simpler pipelines. Each simpler pipeline drops one part of our method: the up and down split, the diffusion smoothing, the high-resolution map, or the model itself. We feed each pipeline the same data and measure the same three errors as Table 8. If our full model beats every simpler version, each part of the method earns its place.

We test three simpler pipelines:

- **Naive.** Copy last week’s counts as the prediction for this week. No model, no smoothing. This is already the dataset-to-dataset row of Table 8 and serves as the noise floor.
- **One-phase PageRank.** A single PageRank chain on the same prior, with no split into an up phase and a down phase. This tests whether the two-phase decomposition of Section 7.1 actually helps.
- **Low-resolution matching.** Keep the full two-phase chain but build the popularity counts from matching on the low-resolution map instead of the high-resolution one. This is the version of the pipeline used before the matching upgrade in Section 3.2, and tests whether the long curvy roads recovered by the high-resolution matcher matter for prediction quality.

Each baseline runs on the same 840 bins as the main evaluation and reports the same three errors.

8.4 Intervention case study: Utah vs Washington, Sept 15 2018

On Saturday September 15, 2018 at 8:00 PM MDT the University of Utah hosted the University of Washington at Rice-Eccles Stadium (40.7596° N, 111.8485° W), with reported attendance of 47,445. The game lies in the middle of our 30-day corpus, so we can ask whether the chain’s pre-kickoff-hour forecast on this particular Saturday differs in a measurable way from a normal Saturday at the same time of day, and whether tuning the chain’s parameters on this event window recovers any of that gap.

Cluster construction. We rank road links by their typical pre-kickoff traffic and keep the busiest few near the stadium. Concretely: take every link whose midpoint lies within 1 km of Rice-Eccles, which leaves 305 candidate links. Each candidate carries a per-bin link probability $E_b(i)$ summing to one over the full network; we

average those per-bin probabilities across the kickoff hour (12 five-minute bins, 19:00 to 19:55) on three clean baseline Saturdays (Sept 1, 8, and 22), giving $12 \times 3 = 36$ probability values per link that we collapse to one scalar “mean kickoff-hour popularity” by averaging. (Sept 15 is the event itself, and Sept 29 is another Utah home football game, so neither is used.) We then take the top- k candidates by that scalar. We report $k \in \{10, 20, 30\}$. The $k = 30$ cluster is shown in Fig. 6: 30 red links concentrated along the east-west arterial spine immediately south of the campus, plus a couple of north-south cross-streets.

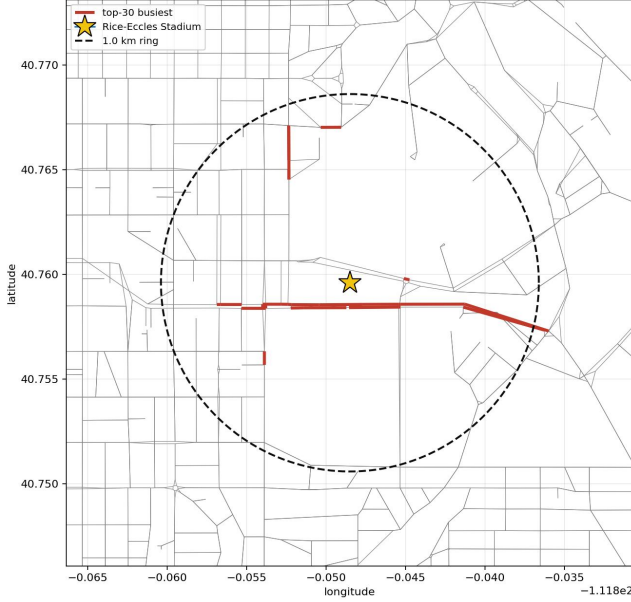


Figure 6: Cluster ranked over 19:00–19:55 on Sept 1, 8, and 22.

Experimental design. The chain’s parameters are trained on non-event Saturday pairs at the same kickoff hour (Sept 1 → Sept 8 and Sept 22 → Sept 29, 19:00–19:55) and then frozen. At test time the frozen chain is given the *trip-origins distribution* of Sept 15, the per-link per-bin count of trips that start on each link during the kickoff hour, normalized to a probability. The origins vector carries the event signal (extra starts around the stadium) without leaking the link-popularity values we are predicting. Parameters never see Sept 15 link densities. The output of the frozen chain on the Sept 15 origins input is compared against Sept 15 link popularity on the top- k cluster; lower is better. Both single-phase and two-phase chains are tuned with the same L-BFGS-B protocol on the training pairs (five multi-start initializations, $p, \beta, \rho \in [0.01, 0.15]$, α_s, α_l unbounded).

Headline result: held-out comparison. Table 11 reports the held-out d2m on the Sept 15 cluster for the proposed two-phase chain, the single-phase one-stage Markov chain baseline, and three simpler predictors that drop one component of the pipeline.

Table 11: Held-out forecast d2m on the Sept 15 stadium cluster. Parameters fit only on the non-event Saturday pairs (Sept 1 → Sept 8 and Sept 22 → Sept 29 at the kickoff hour) and frozen before evaluation on Sept 15. Test input is the Sept 15 trip-origins distribution after γ -diffusion. Lower is better. Read each cell as “this predictor’s mean per-link relative error, averaged over the 12 kickoff-hour bins and the top- k stadium links.”

predictor	top-10	top-20	top-30	notes
two-phase, tuned (ours)	0.678	0.751	0.848	main method
single-phase, tuned	0.709	0.826	0.852	1-stage Markov chain
two-phase, no dangling fix (v1)	1.734	2.160	2.054	earlier two-phase variant
two-phase, no γ -smoothing	1.785	2.249	2.084	raw Laplace prior
d2d (last week’s popularity, no chain)	7.088	8.918	8.062	naive baseline

Reading the table. The two-phase chain is the best predictor on every cluster under the held-out protocol, by 0.031 d2m on top-10, 0.075 d2m on top-20, and 0.004 d2m on top-30 relative to the single-phase chain. The remaining baselines tell a consistent story: the naive d2d row uses last week’s popularity and lands at the event-day noise floor; the v1 chain without the dangling-link fix and the no- γ -smoothing variant both sit well above the tuned chains, showing that the dangling correction and the γ -smoothing component each earn their place in the pipeline.

Table 12: Frozen-tuned parameter values per cluster, fit on the non-event Saturday training pairs (origins → popularity). Parameter ranges: $p, \beta, \rho \in [0.01, 0.15]$, α_s, α_l unbounded. Read each row as “on this cluster size, L-BFGS-B converged to these parameters and produced the d2m bolded in Table 11; α_s, α_l are unbounded exponents and can fall outside $[-1, 1]$.”

cluster	single-phase			two-phase			
	p	α_s	α_l	α_s	α_l	β	ρ
top-10	0.01	−2.39	+4.67	+3.62	−4.22	0.15	0.01
top-20	0.01	−4.11	−0.62	−6.43	−1.03	0.01	0.1049
top-30	0.01	+47.36	−15.03	+77.20	−33.47	0.01	0.01

9 DISCUSSION AND LIMITATIONS

What the held-out experiment shows. Under proper held-out conditions the two-phase chain remains the best predictor on the Sept 15 stadium cluster across the predictors we considered: it beats the single-phase chain, the v1 two-phase chain without the dangling-node fix, the no- γ -smoothing variant, and the no-chain baselines. The gap between the two-phase chain and the single-phase chain is smaller than under the in-sample protocol because the held-out setup denies both chains direct access to the event-day ground truth, but the ordering is preserved across all three cluster sizes (top-10, top-20, top-30). Every component of the pipeline (Laplace pseudo-count, $\gamma = 0.20$ diffusion, dangling-node correction, two-phase split) is shown to earn its place by an ablation against the corresponding simpler variant in Table 11.

Limitations. The probe corpus is a biased sample (Section 3.3): it oversamples freeway corridors and undersamples residential streets. The chain therefore predicts *probe-population link popularity*, not

a uniform-vehicle volume. The validation also tests prediction at a single time scale (5-minute bins) and a single city; cross-city transfer and finer-grained resolutions are outside scope. Finally, the intervention case study uses a single known event; an evaluation across many labeled events would strengthen the generalization claim.

10 CONCLUSION

We presented a two-phase PageRank chain that turns a smoothed per-link popularity prior into a per-link traffic prediction on a 99,716-link city graph, with all calibration steps (σ for map-matching, γ for diffusion smoothing, β and ρ for the chain) driven by the data. Under a proper held-out protocol on a real event (Utah vs Washington, Sept 15 2018), the two-phase chain beats the single-phase baseline and a suite of simpler comparators on every stadium cluster size, validating both the two-phase architecture and the calibration pipeline.

11 CASE STUDY: DETECTING AN UNANNOUNCED EVENT

1. Detection from the ground truth. For every 5-minute slot t we compare the raw map-matched counts at t and at $t + 7$ days, restricted to links carrying at least five trips at t :

$$\text{MRE}(t) = \frac{1}{|S(t)|} \sum_{i \in S(t)} \frac{|c_{t+7}(i) - c_t(i)|}{c_t(i)}, \quad S(t) = \{i : c_t(i) \geq 5\}. \quad (24)$$

Figure 7 plots the per-slot MRE distribution for three consecutive week-pair comparisons. Both comparisons that involve week 2 (W1→W2 and W2→W3) have heavier right tails than W3→W4: slots in week 2 differ from their twins in the adjacent weeks while the W3-versus-W4 comparison is unremarkable. The twelve consecutive 5-minute bins of Tue Sep 11 16:00–17:00 fall in those tails, and this hour is the event window we take into the rest of the case study.

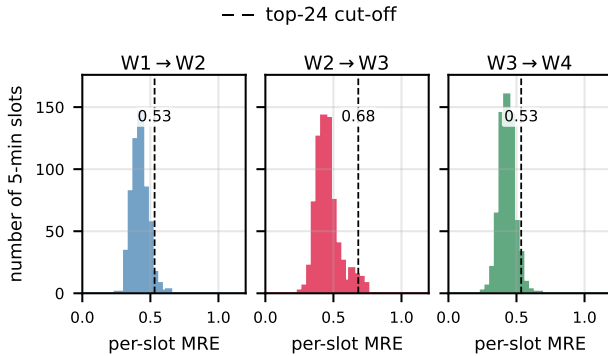


Figure 7: Per-slot MRE for three consecutive week-pair comparisons. W1→W2 and W2→W3 have heavier right tails than W3→W4.

2. External corroboration. With anomalous bins in hand but no idea *why*, we searched local news and government archives for events on Sep 11–13, 2018 in Salt Lake City. The hit is exact: the *Utah State Fair* ran Sep 6–16, 2018 at the Utah State Fairpark, 155 N 1000 West, Salt Lake City, immediately west of I-15 Exit 309 [27]. Its gates opened at noon specifically on Sep 11, 12, and 13 [27], the same three weekdays our top slots fell on. Total attendance was about 283,000, a record-breaking year [28].

3. Focus on twelve consecutive bins. We pick the twelve consecutive 5-minute bins of Tue Sep 11 16:00–17:00 as a representative anomaly hour; they sit inside the Tue-afternoon cluster of the top-24 and the noon-opening day of the Fair. To put the event *on the target side*, we pair them with Tue Sep 4 16:00–17:00 (the same hour one week earlier, before the Fair started on Thu Sep 6). The chain is fed a clean pre-Fair prior and asked to reproduce the Fair-active distribution.

4. Localize the region and pick top-20 links. The news search in step 2 pointed at the Fairpark, served by I-15 Exit 309. Around that exit we take a one-kilometer road-graph neighborhood and keep the twenty links inside it with the highest diffused popularity over the three non-Fair Tuesdays (Sep 4, 18, and 25). Figure 8 confirms the pick on the experiment hour: in each of the twelve 5-minute bins of 16:00–17:00, the Fair Tuesday Sep 11 sits well above the average over the three reference Tuesdays. Hour totals on the twenty-link region are 141 trips in the reference average versus 263 on Sep 11, a 1.87× lift.

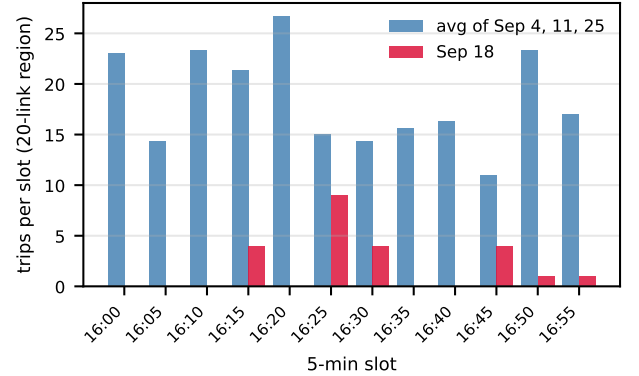


Figure 8: Trips per 5-min bin on the 20-link Fairpark region, 16:00–17:00: reference average over Sep 4, 18, 25 (blue) vs Fair Tue Sep 11 (red).

5. Fit the chain. We tune the classical-form two-phase chain (Appendix A) on the twelve 5-minute pairs (Sep 4 16 : MM, Sep 11 16 : MM), input on the left, target on the right, restricting the loss to the twenty links from step 4. The edge weight gains a length exponent on top of the existing speed and lane exponents:

$$w_{ij} \propto \left(\frac{\text{speed}_j}{\text{speed}} \right)^{\alpha_s} \left(\frac{\text{lanes}_j}{\text{lanes}} \right)^{\alpha_l} \left(\frac{\text{length}_j}{\text{length}} \right)^{\alpha_{\text{len}}}. \quad (25)$$

Bounds: $\alpha, \beta \in [0.01, 0.15]$; $\alpha_s, \alpha_l, \alpha_{\text{len}}$ unbounded. L-BFGS-B with five starts. Table 13 reports the result with and without α_{len} .

Table 13: Classical-form two-phase chain fitted on the 20-link Fairpark region, input Tue Sep 4 16:MM, target Tue Sep 11 16:MM.

configuration	params	Ov. MRE (mean)	Ov. MRE (median)
20-link top-busy ($\alpha, \beta, \alpha_s, \alpha_l$)	4	1.015	0.986
20-link top-busy, $+\alpha_{len}$	5	0.974	0.944

At the best start of the 5-parameter fit, $\alpha_{len} \approx +9.2$, so the chain strongly up-weights long links (arterials and the freeway feeding the Fairpark exit); α and β pin at 0.01 in every start. Adding α_{len} drops the region MRE from 1.015 to 0.974 (−4%). Predicting the Fair distribution from a clean pre-Fair prior is genuinely harder than the reverse direction: the chain has to *create* event-specific structure rather than smooth it away, and the irreducible Overall MRE on this region sits near 1.

REFERENCES

- [1] Wannes Meert and Mathias Verbeke. HMM with non-emitting states for map matching. In *European Conference on Data Analysis (ECDA)*, Paderborn, Germany, July 2018.
- [2] Wannes Meert. *Leuven Map-Matching: a Python library for matching GPS trajectories to road networks*. KU Leuven, DTAI Research Group, 2015–2024. <https://github.com/wannesm/LeuvenMapMatching>.
- [3] Paul Newson and John Krumm. Hidden Markov map matching through noise and sparseness. In *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '09)*, pages 336–343. ACM, 2009. doi:10.1145/1653771.1653818.
- [4] Roger W. Sinnott. Virtues of the Haversine. *Sky and Telescope*, 68:158, 1984. <https://ui.adsabs.harvard.edu/abs/1984S%26T...68..158S/abstract>.
- [5] John P. Snyder. *Map Projections: A Working Manual*. U.S. Geological Survey Professional Paper 1395, U.S. Government Printing Office, Washington, DC, 1987. Chapter 12, “Equidistant Cylindrical Projection”, pp. 90–91. <https://pubs.usgs.gov/pp/1395/report.pdf>.
- [6] Mohammed A. Qudus, Washington Y. Ochieng, and Robert B. Noland. Current map-matching algorithms for transport applications: state-of-the-art and future research directions. *Transportation Research Part C: Emerging Technologies*, 15(5):312–328, 2007. doi:10.1016/j.trc.2007.05.002.
- [7] Yin Lou, Chengyang Zhang, Yu Zheng, Xing Xie, Wei Wang, and Yan Huang. Map-matching for low-sampling-rate GPS trajectories. In *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '09)*, pages 352–361. ACM, 2009. doi:10.1145/1653771.1653820.
- [8] Taras Agryzkov, Leandro Tortosa, and José F. Vicent. Extending the Adapted PageRank Algorithm centrality model for urban street networks using non-local random walks. *Applied Mathematics and Computation*, 447:127531, 2023. doi:10.1016/j.amc.2023.127531.
- [9] Marta C. González, César A. Hidalgo, and Albert-László Barabási. Understanding individual human mobility patterns. *Nature*, 453:779–782, 2008. doi:10.1038/nature06958.
- [10] Dirk Helbing. Traffic and related self-driven many-particle systems. *Reviews of Modern Physics*, 73(4):1067–1141, 2001. doi:10.1103/RevModPhys.73.1067.
- [11] Md. Shahadat Iqbal, Charisma F. Choudhury, Pu Wang, and Marta C. González. Development of origin-destination matrices using mobile phone call data. *Transportation Research Part C: Emerging Technologies*, 40:63–74, 2014. doi:10.1016/j.trc.2014.01.002.
- [12] Xiangdong Yang, Yang Lu, and Wei Hao. Origin-destination estimation using probe vehicle trajectory and link counts. *Journal of Advanced Transportation*, 2017:4341532, 2017. doi:10.1155/2017/4341532.
- [13] Pu Wang, Timothy Hunter, Alexandre M. Bayen, Katja Schechtner, and Marta C. González. Understanding road usage patterns in urban areas. *Scientific Reports*, 2:1001, 2012. doi:10.1038/srep01001.
- [14] Yu Zheng. Trajectory data mining: an overview. *ACM Transactions on Intelligent Systems and Technology*, 6(3):29, 2015. doi:10.1145/2743025.
- [15] Jing Yuan, Yu Zheng, Chengyang Zhang, Wenlei Xie, Xing Xie, Guangzhong Sun, and Yan Huang. T-drive: driving directions based on taxi trajectories. In *Proceedings of the 18th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS '10)*, pages 99–108. ACM, 2010. doi:10.1145/1869790.1869807.
- [16] Daqing Li, Bowen Fu, Yunpeng Wang, Guangquan Lu, Yehiel Berezin, H. Eugene Stanley, and Shlomo Havlin. Percolation transition in dynamical traffic network with evolving critical bottlenecks. *Proceedings of the National Academy of Sciences (PNAS)*, 112(3):669–672, 2015. doi:10.1073/pnas.1419185112.
- [17] Erik Jenelius, Tom Petersen, and Lars-Göran Mattsson. Importance and exposure in road network vulnerability analysis. *Transportation Research Part A: Policy and Practice*, 40(7):537–560, 2006. doi:10.1016/j.tra.2005.11.003.
- [18] Vito Latora and Massimo Marchiori. Vulnerability and protection of infrastructure networks. *Physical Review E*, 71:015103(R), 2005. doi:10.1103/PhysRevE.71.015103.
- [19] David I. Shuman, Sunil K. Narang, Pascal Frossard, Antonio Ortega, and Pierre Vandergheynst. The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains. *IEEE Signal Processing Magazine*, 30(3):83–98, 2013. doi:10.1109/MSP.2012.2235192.
- [20] Michaël Defferrard, Xavier Bresson, and Pierre Vandergheynst. Convolutional neural networks on graphs with fast localized spectral filtering. In *Advances in Neural Information Processing Systems (NeurIPS)* 29, pages 3837–3845, 2016. <https://arxiv.org/abs/1606.09375>.
- [21] Andreas Horni, Kai Nagel, and Kay W. Axhausen (editors). *The Multi-Agent Transport Simulation MATSim*. Ubiquity Press, London, 2016. doi:10.5334/baw.
- [22] Michael Balmer, Kay W. Axhausen, and Kai Nagel. Agent-based demand-modeling framework for large-scale microsimulations. *Transportation Research Record: Journal of the Transportation Research Board*, 1985:125–134, 2006. doi:10.1177/0361198106198500114.
- [23] Minjie Wang, Su Yang, Yi Sun, and Jun Gao. Discovering urban mobility patterns with PageRank based traffic modeling and prediction. *Physica A: Statistical Mechanics and its Applications*, 485:23–34, 2017. doi:10.1016/j.physa.2017.04.155.
- [24] Bin Jiang. Ranking spaces for predicting human movement in an urban environment. *International Journal of Geographical Information Science*, 23(7):823–837, 2009. doi:10.1080/13658810802022822.
- [25] Armando Bazzani, Bruno Giorgini, Sandro Rambaldi, Riccardo Gallotti, and Luca Giovannini. Statistical laws in urban mobility from microscopic GPS data in the area of Florence. *Journal of Statistical Mechanics: Theory and Experiment*, 2010(05):P05001, 2010. doi:10.1088/1742-5468/2010/05/P05001.
- [26] Xiao Liang, Jichang Zhao, Li Dong, and Ke Xu. Unraveling the origin of exponential law in intra-urban human mobility. *Scientific Reports*, 3:2983, 2013. doi:10.1038/srep02983.
- [27] Deseret News staff. Utah State Fair kicks off Sept. 6. *Deseret News*, 22 August 2018. <https://www.deseret.com/2018/8/22/20651762/utah-state-fair-kicks-off-sept-6/>.
- [28] Deseret News staff. Revamped Utah State Fairpark has record-breaking year – but wants \$550K more from state. *Deseret News*, 15 November 2018. Deseret News, 15 Nov 2018.

A APPENDIX: CLASSICAL-FORM TWO-PHASE CHAIN

A.1 Definition

Let P_{cs} be the column-stochastic out-adjacency of the directed road network (each column of P_{cs} sums to one and represents the distribution of next-link probabilities out of a given source link), and let E_b be the diffused popularity prior at the source bin. We build the $2N \times 2N$ block transition

$$M_{2N} = \begin{bmatrix} (1 - \beta) P_{cs} & 0 \\ \beta I & P_{cs} \end{bmatrix}, \quad E_{2N} = \begin{bmatrix} E_b \\ 0 \end{bmatrix}. \quad (26)$$

M_{2N} is column-stochastic, acting directly on probability vectors: from any link in the up phase, a fraction $(1 - \beta)$ of its mass follows the topology within the up phase and a fraction β commits to the same link in the down phase; once in the down phase, mass keeps following the topology with no internal restart. The chain iterates

$$v^{(k+1)} = (1 - \alpha) M_{2N} v^{(k)} + \alpha E_{2N}, \quad (27)$$

where $\alpha \in (0, 1)$ is the global teleportation rate. Component-wise:

$$v_{up}^{(k+1)} = (1 - \alpha)(1 - \beta) P_{cs} v_{up}^{(k)} + \alpha E_b, \quad (28)$$

$$v_{down}^{(k+1)} = (1 - \alpha) \beta v_{up}^{(k)} + (1 - \alpha) P_{cs} v_{down}^{(k)}. \quad (29)$$

Mass lost through dangling nodes (links with no out-edges) is reinjected through E_b at each step so the total $\|v^{(k)}\|_1 = 1$ is preserved. The final prediction is $v = v_{up} + v_{down}$.

A.2 Estimation

The chain is evaluated and tuned on the same 840 calendar (t, t+7 days) pairs as the rest of Table 10, with the same source and target taken from the diffused popularity matrix.

A *tuned* rung fixes $\alpha = 0.15$ and minimizes Overall MRE by gradient descent over $\beta \in [0.01, 0.15]$ together with the road-type exponents α_s, α_l (unbounded), in three modes that mirror the single-phase ladder: lanes-only, speed-only, and both.